물리화학2 (2011년 2학기) 1차 과제 해답 (총 **40점**)

2011년 9월 20일

1. **(총 20 점, 각 10 점)**

(a)

$$\begin{aligned} 1 = \sum_{n=0}^{\infty} P_n(\nu) &= \sum_{n=0}^{\infty} N e^{-nh\nu/(k_B T)} \\ &= N \sum_{n=0}^{\infty} x^n \qquad \left(\text{단, } x = e^{-h\nu/(k_B T)}\right) \\ &= N \frac{1}{1-x} = N \frac{1}{1 - e^{-h\nu/(k_B T)}} \end{aligned}$$

그러므로

$$N = 1 - e^{-h\nu/(k_B T)} \tag{1}$$

(b) (1) 식으로부터

$$P_n(\nu) = (1 - e^{-h\nu/(k_B T)}) e^{-nh\nu/(k_B T)} \tag{2}$$

그러므로

$$\begin{aligned} \langle \epsilon^P_{\text{osc}} \rangle &= \sum_{n=0}^{\infty} nh\nu P_n(\nu) \\ &= \sum_{n=0}^{\infty} nh\nu (1 - e^{-h\nu/(k_B T)}) e^{-nh\nu/(k_B T)} \\ &= h\nu(1-x)x \sum_{n=0}^{\infty} n x^{n-1} \qquad \left(\text{단, } x = e^{-h\nu/(k_B T)}\right) \\ &= n\nu(1-x)x \frac{1}{(1-x)^2} \\ &= n\nu \frac{x}{(1-x)} \\ &= \frac{h\nu e^{-h\nu/(k_B T)}}{1 - e^{-h\nu/(k_B T)}} = \frac{h\nu}{e^{h\nu/(k_B T)} - 1} \end{aligned} \tag{3}$$

2. **(총 20 점, 각 10 점)**

(a)

$$\begin{aligned} 1 = \sum_{n=0}^{\infty} P_n(\nu_E) &= \sum_{n=0}^{\infty} N e^{-(1/2+n)\Theta_E/T} \\ &= N e^{-\Theta_E/(2T)} \sum_{n=0}^{\infty} x^n \qquad \left(\text{단, } x = e^{-\Theta_E/T}\right) \\ &= N e^{-\Theta_E/(2T)} \frac{1}{1-x} = N \frac{e^{-\Theta_E/(2T)}}{1 - e^{-\Theta_E/T}} \end{aligned}$$

그러므로

$$N = \frac{1 - e^{-\Theta_E/T}}{e^{-\Theta_E/(2T)}} \tag{4}$$

(b) (4) 식으로부터

$$\begin{aligned} P_n(\nu) &= \frac{1-e^{-\Theta_E/T}}{e^{-\Theta_E/(2T)}} e^{-(1/2+n)\Theta_E/T} \\ &= (1-e^{-\Theta_E/T}) e^{-n\Theta_E/T} \end{aligned} \tag{5}$$

그러므로

$$\begin{aligned} \langle \epsilon_x^E \rangle &= \sum_{n=0}^{\infty} \left(\frac{1}{2}+n\right) k_B \Theta_E P_n(\nu) \\ &= \frac{1}{2} k_B \Theta_E \sum_{n=0}^{\infty} P_n(\nu_E) + \sum_{n=0}^{\infty} k_B \Theta_E (1-e^{-\Theta_E/T}) e^{-n\Theta_E/T} \\ &= \frac{1}{2} k_B \Theta_E + \sum_{n=0}^{\infty} k_B \Theta_E (1-e^{-\Theta_E/T}) e^{-n\Theta_E/T} \qquad (\sum_{n=0}^{\infty} P_n(\nu_E) = 1 \text{ 이므로}) \\ &= \frac{1}{2} k_B \Theta_E + k_B \Theta_E (1-x) x \sum_{n=0}^{\infty} n x^{n-1} \qquad \left(\text{단, } x = e^{-\Theta_E/T}\right) \\ &= \frac{1}{2} k_B \Theta_E + k_B \theta_E (1-x) x \frac{1}{(1-x)^2} \\ &= \frac{1}{2} k_B \Theta_E + k_B \theta_E \frac{x}{1-x} \\ &= \frac{1}{2} k_B \Theta_E + \frac{k_B \theta_E e^{-\Theta_E/T}}{1-e^{-\Theta_E/T}} \\ &= k_B \left(\frac{\Theta_E}{2} + \frac{\theta_E}{e^{\Theta_E/T}-1} \right) \end{aligned} \tag{6}$$